

CSA1711 - Artificial Intelligence

CO4 Assessment Tool - 2

Article Review - Answer Script

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Task 1: Article Summary

(a) Domain / Application Area and ML Problem Addressed

Domain / Application Area: Healthcare / Biomedical Informatics — clinical diagnosis and management of Diabetes Mellitus using biochemical and demographic patient data.

The specific ML problem addressed is multi-class classification: given 11 clinical/demographic features per patient (blood sugar level, age, gender, creatinine, BMI, urea, cholesterol, LDL, VLDL, triglycerides, HDL, and HbA1c), predict whether the patient belongs to one of three diabetes-status classes — Diabetic (Y), Non-Diabetic (N), or Predicted-Diabetic/borderline (P). This is a supervised inductive learning / decision-tree classification problem, benchmarked against an Artificial Neural Network (ANN), and supplemented with independent statistical analyses (t-tests and Principal Component Analysis) of the same biomarkers.

(b) Key Objective and Research Gap (150-200 words)

The authors set out to determine which of two widely used machine-learning approaches — a Decision Tree Classifier and an Artificial Neural Network — more accurately predicts diabetes status from a real hospital patient dataset, and to go beyond a bare accuracy comparison by also explaining which biochemical indicators drive that prediction and how they differ statistically between diabetic and non-diabetic patients. Much of the existing literature the authors reviewed benchmarks isolated algorithms (Naïve Bayes, SVM, logistic regression, ensemble trees, etc.) on standard datasets such as the Pima Indians dataset, typically reporting only accuracy-type metrics without probing why a model performs well or which biomarkers are clinically meaningful. The gap the authors identify is the lack of a study that (i) directly and rigorously compares a decision tree against an ANN across a full battery of metrics (accuracy, precision, recall, F1, MCC, AUC) on a richer, hospital-sourced biochemical dataset, and (ii) pairs that predictive comparison with statistical hypothesis testing and PCA to add mechanistic, clinically interpretable insight — rather than treating the model as a black box. They fill this gap by combining decision-tree/ANN benchmarking with t-tests and PCA on the same dataset.

Task 2: Methodology Analysis

(a) ML Technique(s) Used and Dataset Description

Techniques used:

- Decision Tree Classifier (CART-style) — split criterion: Gini impurity; hyperparameters: maximum depth 4, minimum 8 samples per leaf, with cost-complexity pruning applied to prevent overfitting.

- Artificial Neural Network (ANN) — a fully connected feedforward network with an input layer of 10 neurons, two hidden layers (6 and 3 neurons respectively), a sigmoid activation function, and an output layer of 2 neurons; hyperparameters were tuned via Random Search.
- Supporting statistical techniques: an independent-samples t-test to compare biomarker means between diabetic and non-diabetic groups, and Principal Component Analysis (PCA) to identify clusters of correlated health indicators.

Dataset:

- Size: 1,000 patient records.
- Source: Patients treated at Al-Kindy Teaching Hospital, Iraq (publicly released via Mendeley Data).
- Features: 11 biochemical/demographic indicators — Age, Gender, Blood Sugar Level, Creatinine (Cr), Urea, BMI, Cholesterol, LDL, VLDL, Triglycerides (TG), HDL, and HbA1c.
- Class distribution: 837 Diabetic (Y), 103 Non-Diabetic (N), 53 Predicted/borderline-Diabetic (P) — a strongly imbalanced multi-class label set.
- Preparation: the dataset was partitioned using an 80:20 hold-out train/test split for both models.

(b) Mapping to CO4 Topics and Strength/Limitation Analysis

This article maps primarily onto the Decision Trees topic of CO4 (a Gini-based, pruned inductive decision tree is the star performer) with the ANN serving as a contrasting non-tree inductive learner, and it also connects to the Statistical Learning Methods topic through its use of hypothesis testing (t-test) and PCA to statistically characterize feature relationships alongside the predictive models. (Reinforcement Learning is not applicable to this article.)

Methodological Strength: The study evaluates both models across a comprehensive set of six metrics (accuracy, precision, recall, F1-score, MCC, and AUC) rather than relying on accuracy alone, and triangulates the predictive comparison with independent statistical validation (t-tests confirming which biomarkers differ significantly between classes, and PCA revealing clusters of physiologically related variables) — giving both a performance verdict and a biologically grounded explanation for it.

Methodological Limitation: The models were evaluated using a single 80:20 hold-out split rather than k-fold cross-validation, and the dataset has a strongly imbalanced class distribution (837 : 103 : 53) with no explicit balancing technique (e.g., SMOTE, class weighting) reported. This combination raises the risk that the very high reported accuracy is partly driven by the dominant 'Diabetic' class and by a single favourable train/test split, rather than being a fully robust, cross-validated estimate of real-world generalization performance.

Task 3: Results & Critical Evaluation

(a) Key Reported Results

Metric	Decision Tree Classifier	Artificial Neural Network (ANN)
Accuracy	97.7%	94.7%
Precision	96.9%	88.8%
Recall (Sensitivity)	96.5%	90.2%
F1-Score	96.5%	90.2%
Matthews Correlation Coefficient (MCC)	87.4%	78.0%
AUC-ROC	96%	78%

The Decision Tree Classifier outperformed the ANN on every reported metric. Feature-importance analysis from the Decision Tree ranked HbA1c as the single most important predictor, followed by BMI and then Age. The independent t-test found statistically significant differences between diabetic and non-diabetic patients for Age, Urea, HbA1c, Cholesterol, Triglycerides, VLDL, and BMI (all $p < 0.05$), while HDL and LDL showed no significant difference between the two groups. PCA identified five interpretable clusters of correlated biomarkers (age-related metabolic indicators, kidney-function markers, cardiovascular lipid profile, lipid-transport/VLDL, and the protective HDL indicator).

(b) Critical Evaluation of the Results

Are the reported metrics realistic? A 97.7% accuracy and 96% AUC for a pruned, shallow (max-depth-4) decision tree are very high for a clinical prediction task, and warrant some scepticism. Two specific concerns stand out:

- Possible label circularity: HbA1c — the model's top-ranked predictor — is itself one of the standard clinical criteria used by physicians to diagnose diabetes in the first place. If the ground-truth Y/N/P labels in this dataset were assigned using HbA1c (or correlated glucose) thresholds, the model may be partly 're-deriving' the diagnostic rule it was trained on rather than discovering an independent predictive relationship, which would inflate the reported accuracy relative to genuine early-detection performance.
- Single train/test split and class imbalance: with only a single 80:20 hold-out split (~200 test patients) and a class distribution as skewed as 837:103:53, the very high headline accuracy could be dominated by the majority 'Diabetic' class, and could vary considerably under a different random split — the absence of k-fold cross-validation with confidence intervals makes it hard to judge how stable these numbers really are.

Potential biases:

- Single-site, single-country dataset (one hospital in Iraq) — the model's performance and feature-importance ranking may not transfer to other populations, ethnicities, or hospital lab-reference ranges without recalibration.
- Gender imbalance in the raw cohort (565 male vs. 435 female) could bias the learned decision boundaries toward patterns more representative of male patients.

Additional experiments that would strengthen the findings:

- k-fold (e.g., 10-fold) cross-validation with reported confidence intervals, instead of a single hold-out split.
- An ablation study that removes HbA1c (and/or blood sugar level) from the feature set, to test whether the model still predicts well using only 'upstream' risk indicators (BMI, age, lipid profile, family history) — this would test for label circularity and better reflect a genuine early-screening use case.
- External validation on an independent dataset from a different hospital/country (e.g., the Pima Indians Diabetes dataset) to test generalizability.
- Explicit class-imbalance handling (e.g., SMOTE or class-weighted training) with before/after comparison, and reporting of balanced accuracy alongside standard accuracy.

(c) Real-World Applicability and Scaling Challenges

Where it can be deployed: a lightweight, interpretable, pruned decision tree of this kind is well suited to integration into hospital Electronic Health Record (EHR) systems as a clinical decision-support flag for at-risk patients during routine biochemical screening, or as a low-compute triage tool in primary-care and low-resource clinical settings that cannot support a heavier ANN pipeline.

Challenges in scaling to industry deployment:

- Data availability: different hospitals/labs use different assay methods and reference ranges for biomarkers like HbA1c and lipid panels, so a model trained on one hospital's data may need local recalibration or retraining before deployment elsewhere.
- Compute: the decision tree itself is computationally cheap to run at inference time; the larger scaling cost is in the surrounding infrastructure — integrating with hospital IT/EHR systems, ensuring data pipelines are HIPAA/GDPR-equivalent compliant, and passing medical-device regulatory review.
- Ethical issues: the dataset is single-site and demographically skewed (gender imbalance, one country), so fairness and equitable performance across diverse patient populations must be validated before wide deployment; clinicians must retain final diagnostic authority (the tool should support, not replace, physician judgement) given the possible HbA1c-circularity concern raised above; and patients should be informed when AI is used to assist in their diagnostic pathway.

Task 4: Reflection & Learning Outcome

(a) Three Key Insights Connected to CO4 Concepts

Insight 1: A carefully pruned, shallow decision tree (max depth 4, cost-complexity pruning) outperformed a more complex ANN on this structured/tabular clinical dataset. This directly reinforces the CO4 concept of pre-pruning vs.

post-pruning in Decision Trees — controlling model complexity to match the data can produce a model that is both more accurate and more interpretable than a more powerful but less-constrained learner, echoing the same 'post-pruned tree generalizes better' conclusion reached in the CO4 Assessment Tool 1 industry-problem task.

Insight 2: The decision tree's feature-importance ranking (HbA1c > BMI > Age) gave a directly interpretable explanation of the model's reasoning, tying back to the CO4 concept of feature-importance analysis in inductive learning: a good classifier should not just be accurate, but should surface a variable-importance ranking that clinicians/domain experts can sanity-check against established medical knowledge.

Insight 3: Pairing the classification model with independent statistical tools — a t-test to confirm which biomarkers genuinely differ between classes, and PCA to reveal latent clusters of correlated variables — connects to the CO4 topic of Statistical Learning Methods, showing that statistical hypothesis testing and dimensionality-reduction techniques can validate and explain a model's behaviour rather than being used only as stand-alone classifiers (e.g., Naïve Bayes/Logistic Regression).

(b) Proposed Extension / Novel Experiment

Proposed experiment: Repeat the Decision-Tree-vs-ANN comparison using 10-fold cross-validation with SMOTE-based class balancing, and add an explicit ablation that removes HbA1c and raw blood-sugar level from the feature set — then re-evaluate accuracy/AUC using only 'upstream' risk indicators (BMI, age, lipid profile, urea/creatinine, family history if available) — with external validation on an independent dataset (e.g., the Pima Indians Diabetes dataset or a different hospital's cohort).

Justification of feasibility:

- All required components are readily available and low-cost: k-fold cross-validation and SMOTE are standard, off-the-shelf functions in libraries such as scikit-learn and imbalanced-learn; a decision tree is computationally inexpensive to retrain many times across folds and ablation settings; and suitable external/open datasets (Pima Indians Diabetes, other Mendeley-hosted diabetes cohorts) already exist for cross-population validation.

Expected impact:

- The cross-validation + SMOTE step would produce more statistically defensible, imbalance-robust performance estimates than the current single hold-out split.
- The HbA1c/blood-sugar ablation would directly test the label-circularity concern raised in Task 3(b); if the model still performs well without these near-diagnostic features, it becomes genuinely useful for early, pre-diagnostic risk screening (catching at-risk patients before a formal HbA1c-based diagnosis) — a materially higher-value clinical use case than confirmatory classification.
- External validation on a second population would directly address the single-site generalizability and demographic-bias concerns identified in Task 3(c), strengthening the case for real-world, multi-hospital deployment.